



BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: ParkMate - A Robotic Assistant for Parking Cars by

[Group_06]

[Section 04]

Group Members:

Rifat Mahmud Tamim	21201787
Faishal Monir	22101235
Rubiya Karim	22101705
Rajin Ibna Rajuanur Rahman	22101717
Umma Salma Mim	22101870

Department of Computer Science and Engineering

BRAC University

05-01-2026

Abstract

The problem of parking has become a significant issue in congested urban areas because of the growing number of vehicles in the same space. ParkMate is an automated parking aid that is aimed at automating the process of vehicle access, parking, slot management, electric charging system, area cleaning, and vehicle exit without any human help. The system identifies incoming cars, opens and closes entry and exit gates automatically, directs the drivers to the parking spaces with visual aids, and detects the occupancy of the slots in real-time. Moreover, ParkMate will have electric vehicle (EV) charging support and automatic slot-cleaning to make the usage process more convenient and efficient in terms of maintenance. The system also eliminates most human intervention through a main controller based on an Arduino UNO, ESP 8266, Ultrasonic Sonar sensors, IR sensors, a DC motor, and display units, thereby enhancing efficiency, safety, and organization. This project will explore how robotics and embedded systems can be practically used to solve a real problem of parking.

Keywords

Microcontroller, Sensors, Automation, Smart parking, Sustainability

1. Introduction

The high rate of urbanization and increase in vehicles have necessitated effective management of parking in recent times. The drivers usually spend a lot of time seeking an available parking space, which adds to the traffic congestion, the amount of fuel used, and emissions. The conventional parking systems are not monitored in real time, and they are highly dependent on human monitoring, which causes inefficiencies and errors. ParkMate aimed to minimize this issue by pointing out these problems and making an automated parking system.

Parking systems that are manual are inefficient, time-wasting, and error possibility. They also do not make real-time information available on the available parking slots, and they do not have automated access control or accommodate modern needs, including EV charging and automated maintenance. Also, manual parking is not as efficient as an automated parking system like ParkMate.

This project is of great significance to Computer Interfacing since it combines microcontrollers with various sensors, actuators, and a display module to form a full embedded robotic system.

2. Related Work / Inspiration

Several smart parking systems have been suggested in recent years based on the use of IoT, sensor networks, and cloud-based monitoring. The current solutions are normally mobile application-based with centralized data processing, but use costly infrastructure and are complicated to implement.

ParkMate is based on these systems but differentiates itself by offering a low- cost stand-alone robotic system that is applicable in small to medium-sized parking. ParkMate, unlike most of the systems it claims to compensate, also incorporates EV charging functions and an automatic slot cleaning system, which will help in terms of both sustainability and maintainability.

3. Technical Approach

3.1 System Architecture

ParkMate has three main layers:

- Input layer: Entrance sensor, Exit sensor, IR Sensors, and Slot Sensors.
- Processing layer: Arduino UNO reads sensors and makes decisions, also transfer data to the ESP8266.
- Output layer: Gate Servo Motor, Slots Servo Motor, Slot DC Motor, 5V Relay Module, RGB LEDs, and LCD Display.

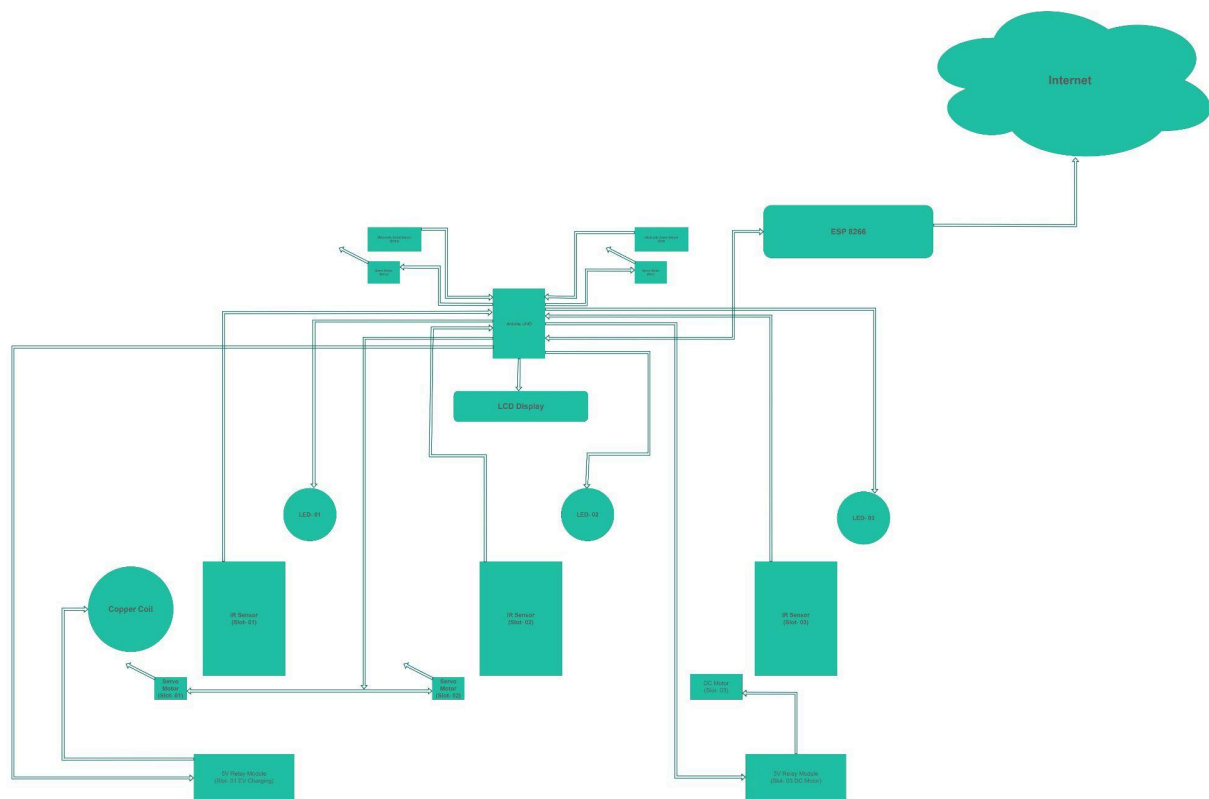


Figure 1: Block diagram of ParkMate (three-slot prototype).

ParkMate System Flowchart

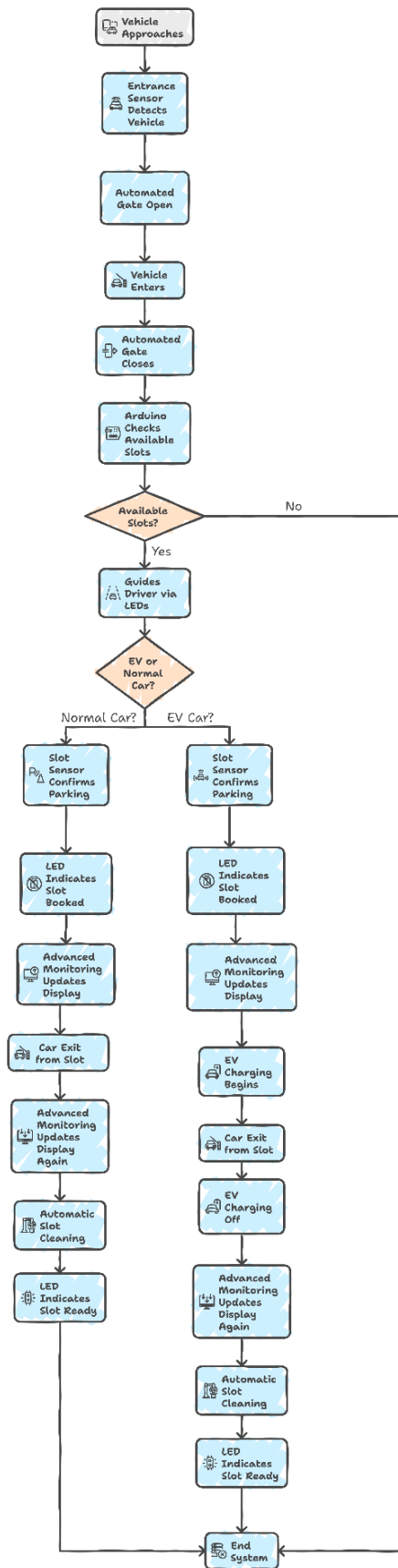


Figure 2: Workflow flowchart for ParkMate.

3.2 Components Used

Hardware Components:

- Arduino UNO is the main controller.
- Ultrasonic Sonar sensor (HC-SR04) to detect a car at the entrance and exit.
- ESP 8266 for Automated Dashboard Control through the Internet.
- Slot Car detection IR sensors (1 Sensor per slot)
- Servo motor (SG90) used for cleaning the floor in slot- 1 & 2.
- DC motor used for cleaning the floor in slot- 3.
- LEDs denoting the slot status (green: empty, off: occupied).
- LCD to show the number of cars and free parking spaces.
- Charging system for EV Vehicles in slot- 1.
- Resistors, diode, transistors, 5V relay module, wires, breadboard, and base platform.

Software:

- Arduino IDE (C/C++) as a means of reading sensors and controlling motors, displays, and LEDs. ESP 8266 for Automated Dashboard Control (Using Google Spreadsheet) through the Internet.

3.3 Cost Breakdown

Table 1 shows the estimated component cost.

Components	Unit Cost	Total (BDT)
Arduino UNO	1 x 1000	1000
Ultrasonic Sensor (HC-SR04)	2 x 100	200
IR Sensor Module	3 x 45	135
RGB LEDs	5 x 10	50
LCD Display	1 x 340	340
Small Servo Motor (SG90)	5 x 130	650
DC Motor (Mini)	1 x 85	85
Power Supply (5V/12V DC Adapter)	1 x 600	600

Jumper Wires /	1 x 100	100
Breadboard	1 x 100	100
Acrylic Sheet / Plywood Base	1 x 50	50
Resistors	5 x 5	25

Table 1: Cost breakdown (BDT).

3.4 Schematic

The circuit will be constructed by attaching sensors to Arduino digital pins and supplying the servo and sensors with the stable 5V. In our design, the output of the IR sensors is attached to an individual Arduino pin. The servo signal pin is attached to a PWM pin. The LEDs are attached with resistors to safeguard them. Grounds are linked to each other (common ground).

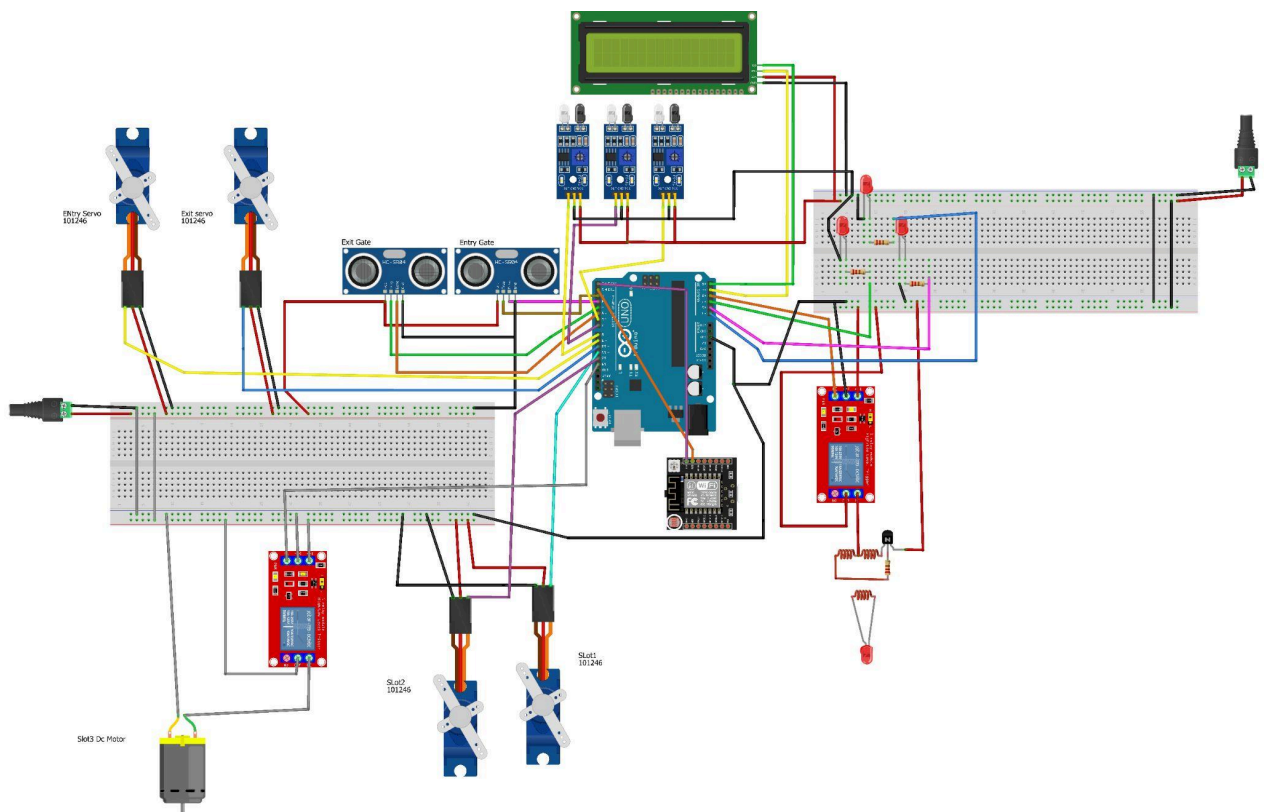


Figure 3: Circuit Diagram for ParkMate.

3.5 Functionality

Step-by-step workflow:

- 1) The sensor in the entry gate will detect a car within the vicinity of the gate.
- 2) The Arduino opens the gate using the servo motor.
- 3) The Arduino checks the available slots through IR sensors from three slots.
- 4) The Arduino activates the RGB light of the vacant slot to green, and the already occupied slot shows green light off.
- 5) On insertion of a vehicle in a slot, the slot sensor changes to a different value, and the Arduino turns the green LED off.
- 6) The number of cars and free slots is displayed in LCD Display.
- 7) EV can charge on slot 1 (Only Slot- 1 provides EV Charging support).
- 8) The automated dashboard system will display all the reports about the parking system.
- 9) Exit gate will control the smooth exit process of cars, same as entry gate.

3.6 Challenges

Major problems were sensor calibration, motor power control, and the coordination of control of various components. These concerns were addressed by thorough testing and correct voltage regulation, as well as software component design. The biggest challenge during the build and testing process is sensor placement and constant power in most cases. IR sensors may provide false results in the event that they are misplaced or lighting changes. Angles and reflections can influence ultrasonic sensors.

4. Sustainability & Impact

4.1 Sustainability

ParkMate enhances environmental sustainability by minimizing the movement of vehicles in the parking premises, which minimizes fuel and emissions. The system advocates EV charging and has energy-efficient elements. The potential of ParkMate is that it will save on unnecessary traffic in a car park, as ParkMate indicates the slots available within a very short time. This

would help to save time and minimize the scale congestion at parking lots. Low-power electronics (LEDs and sensors) are also contained in the prototype, thus, low energy consumption. The design of the system was also to be modular, and hence, the components can be reused or repaired rather than the entire system being replaced.

4.2 Impact

ParkMate is applied to enhance the user experience since it makes the parking process simpler and the stressful. It can ensure that the time one spends at the entrance will decrease as the gate will open automatically. In a clear slot indication, drivers decide on the appropriate space within a short duration of time. The same concept can be replicated in the future to a larger parking space and additional sensors, and a stronger control unit.

4.3 Future Work

The same would be developed by installing an exit sensor that would result in the number of cars being more exact, a small display sign that would indicate the slot that is available specifically, and installing a wireless monitoring system so that the car can be viewed remotely by the user. With respect to the EV characteristics, a low-voltage test can be made safe and expanded into a full-size prototype of a real charging management system.

4.4 Limitations

It is a miniature version that comes with three vacancies, hence lacks many of the characteristics of the actual world including a weather shield, massive motor drivers, and a robust mechanical shield. The sensor-based readings could change with location, surface reflectiveness, and lighting. Also, the EV charging and cleaning infrastructure is hypothetical on this project, and it would need more safety and hardware engineering to be implemented in the field.

5. Results & Discussion

The viability tests of the essential functions of ParkMate were carried out: entrance detection, gates opening/closing, three slots slot slot detection, LED update, and display update. The right positioning of sensors and the constant power are the main aspects that characterize the system's performance. Table 3 provides a blank testing paper on which a person might record the observations and results of the demonstrations.

Test Item	What to Check	Pass/Fail
Entrance & Exit detection	Car within a specified distance.	Pass
Gate control	The servo operates in a smooth manner.	Pass
Slot 1 detection	The IR sensor varies with the car being placed and removed.	Pass
Slot 2 detection	The IR sensor varies with the car being placed and removed.	Pass
Slot 3 detection	The IR sensor varies with the car being placed and removed.	Pass
Slot 1 EV	A wireless charging system is used.	Pass
LED guidance	Empty-green, occupied-red	Pass

Table 2: Testing sheet

6. Conclusion

ParkMate is a smart parking prototype that is a three-slot parking device that employs sensors, an Arduino UNO, and actuators to automate the guidance of parking and control of a gate. It reads a car coming in and opens the gate, turns on a display that allows it to count the number of cars and the spaces that are open with the use of LEDs. The project illustrates fundamental concepts of robotics, including sensing, decision-making, and actuation. As it continues to be improved, the same concept can be expanded to accommodate more parking spaces and additional options such as EV control and automated cleaning.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions (summary)
Rifat Mahmud Tamim	21201787	Hardware design (Partial), Wiring sensors	Assembled IR sensors and Servo motor wiring; assisted in assembling.
Faishal Monir	22101235	Software Design, Control Logic, Automated Dashboard Design, EV Charging System Design	Build software related structure and coding for specific implementation.
Rubiya Karim	22101705	Documentation and Report prepared, Tested Wifi Connection Mechanism through ESP 8266	Test recording, prepared report, Tested Wifi module design for data transfer.
Rajin Ibna Rajuanur Rahman	22101717	Display and LEDs Control, DC Motor functionality, Full Poster Prepared, Logistic Supply	Tested full system and supported LCD display and LEDs connections. Build DC Motor functionality as floor cleaning actuator. Prepared all the diagrams and one poster for the documentation.
Umma Salma Mim	22101870	Mechanical setup, Build Hardware structure, Data	Tested EV Charging System with Charging Devices. Project Design - Layout of designs in

		validation and Maintenance of the Project	gates and the base.
--	--	---	---------------------

Table 4: Contribution table

References

Xiao, R.I., Jaller, M. Prediction framework for parking search cruising time and emissions in dense urban areas. *Transportation* 52, 1289–1317 (2025).

<https://doi.org/10.1007/s11116-023-10455-4>